

Izabella Faria

Livia Coutinho

Mariana Silva

Mateus Neves

[Dynamic pricing model for advertising space on Rádio Gazeta]

SÃO PAULO
2025

Izabella Faria

Livia Coutinho

Mariana Silva

Mateus Neves

[Dynamic pricing model for advertising space on Rádio Gazeta]

Final Course Project submitted to the
Institute of Technology and Leadership
(INTELI), to obtain a bachelor's degree in
[Software Engineering] [Information
Systems].

Advisor: Prof. [Ovídio Lopes]



SÃO PAULO
2025

Cataloging in Publication
Library and Documentation Service
Institute of Technology and Leadership (INTELI)
Data entered by the author.

Sobrenome, Nome

Título do trabalho: subtítulo / Nome Sobrenome do autor; Nome e Sobrenome do orientador. – São Paulo, 2025.

nº de páginas : il.

Trabalho de Conclusão de Curso (Graduação) – Curso de [Ciência da Computação] [Engenharia de Software] [Engenharia de Hardware] [Sistema de Informação] / Instituto de Tecnologia e Liderança.

Bibliografia

1. [Assunto A]. 2. [Assunto B]. 3. [Assunto C].

CDD. 23. ed.

Acknowledgments

The authors would like to express their sincere gratitude to **Inteli – Institute of Technology and Leadership** for providing an academic environment that fosters innovation, critical thinking, and real-world problem solving.

We also thank our academic supervisor, **Professor Ovidio Lopes**, for his guidance, technical support, and valuable insights throughout the development of this project.

Finally, we would like to thank **Rede Gazeta** for the partnership, trust, and for providing the business context and data that made this project possible

Resumo

[Faria, Izabella; Coutinho, Livia; Silva, Mariana; Neves, Mateus. **Modelo de precificação dinâmica para espaços publicitários da Rádio Gazeta**. 2025. 25 páginas. TCC (Graduação) – Curso [Engenharia de Software] [Sistema de Informação], Instituto de Tecnologia e Liderança, São Paulo, 2025.]

[O presente trabalho apresenta o desenvolvimento e a implementação de um modelo de precificação dinâmica para os espaços publicitários da Rádio Gazeta, com o objetivo de otimizar a receita e modernizar a gestão de inventário. Diante de um cenário de tabelas estáticas e negociações manuais, o projeto propôs uma solução baseada em inteligência de dados para capturar a sensibilidade da demanda. A metodologia integrou uma análise exploratória dos dados históricos de vendas com técnicas de aprendizado de máquina para previsão de preços e, centralmente, desenvolveu um modelo econométrico de elasticidade-preço (*log-log*) para classificar o comportamento de compra dos clientes. Como resultado final, obteve-se uma ferramenta prescritiva capaz de diagnosticar a elasticidade por produto e horário, gerando recomendações de faixas de preço ótimas e estratégias de ajuste (manter, aumentar ou reduzir valores) conforme a taxa de ocupação. Conclui-se que a solução entrega à Rádio Gazeta um mecanismo de suporte à decisão, permitindo a transição de uma precificação tabelada para uma abordagem estratégica e orientada a dados, que busca maximizar a rentabilidade e o volume de vendas.].

Palavras-Chave: [precificação dinâmica]; [elasticidade de preço]; [inteligência de dados]; [aprendizado de máquina]; [publicidade em rádio].

ABSTRACT

[Faria, Izabella; Coutinho, Livia; Silva, Mariana; Neves, Mateus. **Dynamic pricing model for Rádio Gazeta advertising spaces**. 2025. 25 pages. Final course project (Bachelor) – Course [Software Engineering] [Information Systems], Institute of Technology and Leadership, São Paulo, 2025.]

[This work presents the development and implementation of a dynamic pricing model for Rádio Gazeta's advertising spaces, aiming to optimize revenue and modernize inventory management. Faced with a scenario of static rate cards and manual negotiations, the project proposed a solution based on data intelligence to capture demand sensitivity. The methodology integrated an exploratory analysis of historical sales data with machine learning techniques for price prediction and, centrally, developed an econometric price elasticity model (log-log) to classify customer purchasing behavior. As a final result, a prescriptive tool was obtained capable of diagnosing elasticity by product and time slot, generating recommendations for optimal price ranges and adjustment strategies (maintain, increase, or reduce values) according to the occupancy rate. It is concluded that the solution provides Rádio Gazeta with a decision support mechanism, enabling the transition from rate-card pricing to a strategic, data-driven approach seeking to maximize profitability and sales volume.]

Key words: [dynamic pricing]; [price elasticity]; [data intelligence]; [machine learning]; [radio advertising].

Summary

1	Introduction	10
2	[Solution Development]	11
2.1	[Applied Rationale]	11
2.2	[Specification and Development	12]
2.3	[Assessment of Impact and Contribution to the Business]	13
3	[Conclusion]	14
	References	14
	Appendices	16
	Annexes	16

1. Introduction

The broadcasting industry operates under the logic of perishable goods, in which inventory—namely, the available airtime for advertising—has a fixed capacity and becomes worthless if not sold at the appropriate moment. Historically, the management of this asset at Rádio Litoral Vitória (Rede Gazeta) has relied on static pricing tables and manual negotiation processes, where the final selling price depends largely on the tacit knowledge of the sales team and on reactive adjustments to market pressures. Although this traditional model has sustained operations to date, it presents inherent limitations, as it fails to capture in real time the stochastic nature of demand, the heterogeneity of customers' willingness to pay, and intra-annual and intra-week seasonal patterns.

In this context, the persistence of rigid prices in the face of dynamic demand leads to economic inefficiencies known as deadweight loss: during periods of high demand, the broadcaster fails to capture consumer surplus (underpricing), while in periods of low demand it faces inventory idle time due to the absence of adequate price incentives (overpricing). The transition toward a Dynamic Pricing and Revenue Management approach therefore emerges not merely as a technological upgrade, but as a strategic imperative for the organization's financial sustainability and competitiveness.

This project, developed within the Corporate track, proposes to overcome the current deterministic model through the design of a data-driven analytical architecture. The solution is based on the integration of Machine Learning methods for market value prediction with econometric modeling to estimate the price elasticity of demand. By transforming historical transactional data into prescriptive intelligence, the project aims to equip Rádio Gazeta with quantitative decision-making tools, enabling agile adjustments to pricing policies according to inventory utilization rates and the estimated sensitivity of each customer segment.

1.1. Partner company context:

Rede Gazeta, founded in 1928, has established itself as the largest media group in the state of Espírito Santo, integrating operations across television (Globo

affiliates), radio, print, and digital media. The organization plays a central role in the region's socioeconomic and cultural development, supported by pillars of journalistic credibility and commercial ethics. The scope of this project is limited to the Radio business unit, with Rádio Litoral Vitória—an audience-leading station and a strategic asset in the group's portfolio—serving as the object of study.

The current advertising market presents significant challenges for traditional media outlets. The rise of digital platforms (AdTech companies and social media networks) has introduced new media-buying paradigms, characterized by real-time performance metrics, granular targeting, and algorithmic pricing. To remain competitive in the face of these new entrants and to sustain radio's relevance as a mass medium, Rede Gazeta has identified a strategic need to modernize its commercial management. This project is therefore embedded in a broader corporate digital transformation initiative, aiming to shift from an empirically driven culture to a data-driven management approach, optimizing the profitability of its inherently limited physical assets.

1.2. Problem definition:

The central problem examined in this study lies in the allocative inefficiency arising from the rigidity of traditional pricing models when confronted with the stochastic volatility of advertising demand. At Rádio Litoral Vitória, the price formation mechanism is based on nominal rate cards which, due to their static nature, fail to reflect the momentary equilibrium between supply and demand. As a result, the final transaction value (net price) is established through decentralized, manual negotiation processes, introducing non-systematic variability and subjectivity into revenue management.

This misalignment between the pricing model and market dynamics leads to operational and financial inefficiencies that can be structured along three critical dimensions:

- **Marginal Revenue (Yield) Suboptimization:** The absence of demand-elasticity-sensitive mechanisms prevents efficient price discrimination. During periods of high seasonality or peak-time slots (prime time), the broadcaster maintains standardized historical discounts, foregoing

the capture of consumer surplus and constraining revenue maximization per unit of airtime sold.

- **Opportunity Cost Due to Idle Capacity (Perishability):** Given the perishable nature of broadcasting inventory, the inability to dynamically adjust prices in low-demand scenarios results in unsold advertising slots (idle schedules). Rate card rigidity inhibits the application of calculated price incentives that could stimulate sales volume and contribute to fixed-cost coverage, leading to wasted productive capacity.
- **Decision-Making Latency and Cognitive Bias:** The current negotiation process relies excessively on the tacit knowledge and individual heuristics of commercial managers. The lack of data-driven analytical support introduces cognitive biases in discounting decisions and prolongs the sales cycle, hindering strategic standardization and weakening the technical defense of the product's value in an increasingly metrics-driven advertising environment.

1.3. Proposed solution and expected contribution:

This study proposes the development of a Decision Support System (DSS) for dynamic pricing, designed to function as an analytical intelligence layer that complements the commercial expertise of Rede Gazeta's sales team. The solution is not intended to replace human negotiation, but rather to equip it with robust quantitative data, enabling managers' market intuition to be validated and enhanced by statistical evidence.

The system architecture processes the broadcaster's transactional history to deliver tactical recommendations and is structured around two main analytical pillars:

- **Econometric Modeling of Price Elasticity:** Application of log-linear regression (log-log model) to estimate demand elasticity coefficients (β). This step translates advertisers' historical behavior into quantifiable metrics, identifying which inventory segments are sensitive to price changes (elastic) and which allow for greater margin protection (inelastic).
- **Prescriptive Decision Matrix (Human-in-the-Loop):** Implementation of a rules-based engine that leverages the diagnosed elasticity to suggest negotiation ranges (floor and ceiling prices). The system generates pricing guidelines that serve as a starting point for negotiation:

- For elastic products, greater price flexibility is recommended to stimulate sales volume.
- For inelastic products, maintaining or increasing prices is recommended to maximize margins.

The expected contribution is the delivery of a tool that reduces information asymmetry and uncertainty in the sales process. By providing pricing parameters grounded in microeconomic theory, the project seeks to optimize allocative efficiency, enabling Rede Gazeta to maximize financial performance through the synergy between the tacit knowledge of its team and the model's analytical processing capabilities.

1.4. Business Objectives:

The strategic alignment of this project with Rede Gazeta's corporate guidelines is intended to deliver tangible gains in competitiveness and efficiency:

- **Revenue Yield Maximization:** Increase overall revenue by capturing price premiums in inelastic inventory segments (high-demand scenarios), thereby mitigating margin erosion.
- **Sales Volume Stimulation:** Identify opportunities within elastic segments where calculated price incentives can increase the number of advertising insertions sold, reducing inventory waste.
- **Pricing Governance and Standardization:** Reduce information asymmetry and negotiation subjectivity by providing the sales team with statistically validated pricing parameters (floor and ceiling) to guide each negotiation.

1.5. Structure of the dissertation:

Chapter 2:

- **Solution Development:** Presents the Applied Rationale, discussing the technologies and methods that justify the technical choice within the business context. The chapter covers the rationale from the Business Area (industry concepts and benchmarking) and the Technological Rationale (concept review and solution justification), in addition to substantiating the management and development methods (such as Agile and PMBOK) used in the corporate environment.
- **Specification and Development:** Details the technical specifications of the solution aligned with the partner company's standards. Presents the Requirements and Specifications (functional, non-functional, and use cases), the Architecture and Technology (integration into the IT ecosystem), the MVP Development and Implementation process (methodology, deployment), and the Technical Assessment (acceptance, performance, and security testing strategies).
- **Impact Assessment and Business Contribution:** Measures the return on investment and the solution's effectiveness. Defines the Corporate Success Metrics (KPIs and measurement methodology), presents the Results and Impact Analysis (comparison with the baseline, quantitative and qualitative analysis), performs the Cost-Benefit Analysis (cost estimation, ROI), and discusses the Critical Success Factors and Lessons Learned.

Chapter 3:

- **Conclusion:** Discusses the achievement of the project's established objectives, addresses the generated business impacts, and presents recommendations for the solution's evolution (new features, scalability, and maintenance), including the knowledge transfer plan and technical documentation for Rede Gazeta's internal team.

2 [Solution Development]

This chapter presents the applied rationale behind the development of the proposed solution, detailing the business foundations, technological choices, and analytical methodologies that support the dynamic pricing model designed for Rede Gazeta. The objective is to demonstrate how business needs, data availability, and analytical techniques were integrated into a coherent and scalable solution.

2.1 [Applied Rationale]

The construction of the solution emerged from an integrated analysis that combined challenges specific to the broadcasting and media advertising sector with the analytical capabilities provided by Data Science and Econometrics. Rather than focusing solely on predictive accuracy, the project prioritized interpretability, operational applicability, and strategic decision support.

The adopted approach reflects the understanding that pricing, in this context, is not merely an operational parameter, but a strategic lever capable of influencing both revenue and inventory utilization. Therefore, the solution was designed to diagnose demand behavior and prescribe pricing actions aligned with real market dynamics.

2.1.1 Business Area Rationale:

From a business perspective, Rede Gazeta operates in a market characterized by fixed inventory, perishability of advertising slots, and strong demand seasonality. Advertising spaces that are not sold within their respective time windows represent irreversible revenue loss, making pricing decisions critical for both occupancy and profitability.

The traditional pricing model used by the company is based on fixed price tables, which treat periods of high and low demand similarly. This rigidity leads to two major inefficiencies:

- **Underutilization of inventory during low-demand periods**, and
- **Loss of margin during high-demand periods**, when prices fail to capture customers' willingness to pay.

Benchmarking against best practices in industries with similar characteristics—such as airlines, hospitality, and digital advertising—reveals that dynamic pricing and demand-sensitive strategies are widely adopted to address these challenges. These industries rely on elasticity analysis to understand how demand reacts to price changes and to guide pricing adjustments accordingly.

In this context, applying price elasticity concepts to radio advertising aligns with market best practices, allowing Rede Gazeta to move from static pricing toward a more adaptive and revenue-oriented strategy.

2.1.2 Technological rationale for the solution:

The technological foundation of the solution was designed to balance analytical robustness, transparency, and ease of integration with the partner company's environment.

All analytical development was conducted using **Python**, leveraging its strong ecosystem for data analysis, econometric modeling, and reproducibility. The use of **Google Colab** enabled collaborative development, version control, and computational scalability without imposing infrastructure constraints on the company.

The solution processes historical data provided by Rede Gazeta in **Excel format**, stored in **Google Drive**, which facilitated secure data sharing and controlled access. These datasets were subjected to structured **ETL (Extract, Transform, Load)** pipelines, exploratory data analysis, and data quality checks before model development.

From a methodological standpoint, the choice of **econometric regression models**—particularly elasticity-based models—was intentional. Unlike black-box machine learning approaches, elasticity models provide:

- Clear economic interpretation (price–demand relationship),
- Actionable insights (increase, reduce, or maintain price),
- Transparency and explainability, which are critical for adoption by business stakeholders.

The final elasticity model operates at a granular level (product × month), producing prescriptive outputs that are later transformed into structured JSON files. These outputs are consumed by the frontend interface, enabling visualization and simulation without manual data input.

From an architectural perspective, while the current prototype feeds the frontend through static data files, the recommended evolution is the integration with Rede Gazeta's backend infrastructure. In a production environment, the model would run periodically (e.g., daily or monthly), ingest updated data directly from internal databases, and automatically publish updated pricing recommendations to the frontend via APIs.

This technological design ensures scalability, maintainability, and alignment with enterprise-grade data workflows.

2.1.3 Fundamentals of Management and Development Methods:

The development of this project followed modern management and software development principles, aligned with practices commonly adopted in corporate environments.

In terms of development practices, the project aligns conceptually with **DevOps principles**, even though a full CI/CD pipeline was not implemented at this stage. Automation of data processing pipelines, reproducibility of experiments, and clear separation between data preparation, modeling, and presentation layers were treated as foundational practices, preparing the solution for future productionization.

2.2 [Specification and Development :]

This section details the technical specifications of the proposed solution, ensuring alignment with Rede Gazeta's operational needs, infrastructure constraints, and long-term scalability goals.

2.2.1 Requirements and Specifications:

Functional Requirements

- Diagnose demand behavior for advertising inventory based on historical data.
- Classify pricing scenarios as **elastic**, **inelastic**, or **inconclusive**.
- Generate recommended price ranges per product, format, and time period.
- Support scenario simulation to evaluate the impact of price changes on revenue and occupancy.
- Provide structured outputs (JSON) to feed the frontend visualization layer.

Non-Functional Requirements

- Data confidentiality and compliance with LGPD principles.
- Reproducibility of results and transparency of model logic.
- Scalability to support additional radios, formats, and time horizons.
- Maintainability of data pipelines and models.
- Compatibility with Rede Gazeta's existing IT ecosystem.

User Specifications and Use Cases

- **Commercial Team:** Use price recommendations and scenario simulations to support negotiations.
- **Management:** Analyze pricing strategy performance and revenue optimization opportunities.
- **Technical Teams:** Integrate model outputs into internal systems and dashboards.

2.2.2 Architecture and Technology:

System Architecture

The proposed solution follows a **modular client-server architecture**, composed of:

- **Data & Modeling Layer:**
Implemented entirely in **Python**, using Google Colab as the development environment. This layer is responsible for ETL, exploratory analysis, modeling, and result generation.

- **Backend Layer (Recommended):**

A future backend service responsible for executing the models on updated data and exposing results via APIs.

- **Frontend Layer:**

A web-based interface that consumes structured JSON outputs to visualize price recommendations and diagnostic insights.

Integration into the IT Ecosystem

In the current MVP, frontend data is loaded from static files. For production, it is recommended that:

- The pricing and elasticity models run automatically using data sourced directly from Rede Gazeta's databases.
- The backend generates updated outputs periodically (daily or monthly).
- The frontend consumes these outputs dynamically via authenticated APIs, eliminating manual data handling.

2.2.3 Development and Implementation (MVP):

The MVP was developed using an **iterative and exploratory approach**, combining Agile practices with data science workflows.

Key characteristics of the MVP include:

- End-to-end pipelines for data ingestion, cleaning, feature engineering, modeling, and output generation.
- Incremental model experimentation (structural, seasonal, negotiation, historical).
- A final elasticity-based model that generates actionable pricing recommendations.
- A frontend prototype demonstrating how results can be consumed and used by business users.

The MVP serves as a proof of concept, validating both technical feasibility and business relevance.

2.2.4 Testing and Technical Evaluation:

Given the analytical nature of the project, testing focused primarily on **model validation and consistency**, rather than traditional software testing alone.

Key evaluation strategies included:

- **Statistical Validation:**

Use of metrics such as R^2 and RMSE to evaluate model performance.

- **Comparative Experiments:**

Incremental model construction to isolate the impact of each group of variables.

- **Scenario Analysis:**

Simulation of price variations to assess revenue sensitivity and robustness.

- **User-Oriented Validation:**

Assessment of interpretability and usefulness of outputs for decision-making.

Future production deployments may include:

- User Acceptance Testing (UAT).
- Performance testing for large-scale data processing.
- Security and access control testing.

2.3 [Assessment of Impact and Contribution to the Business]

This section evaluates the expected business impact of the proposed dynamic pricing solution, considering the return on investment of time, analytical effort, and technological resources. Given that the project was developed as a pilot and proof of concept, the assessment focuses on measurable indicators, potential financial gains, and qualitative improvements in decision-making and operational efficiency.

2.3.1 Defining Corporate Success Metrics:

To evaluate the success of the solution, a set of business-oriented Key Performance Indicators (KPIs) was defined in alignment with Rede Gazeta's revenue and inventory management objectives.

Project KPIs

The main indicators considered include:

- **Advertising Revenue (R\$):** total revenue generated from commercial insertions.
- **Inventory Occupancy Rate (%):** proportion of available advertising inventory effectively sold.
- **Average Price per Second (R\$/s):** normalized metric to compare prices across formats and durations.
- **Discount Rate (%):** difference between table price and final negotiated price.
- **Price Variability Index:** degree of flexibility applied to pricing across periods of different demand.

Measurement Methodology

The evaluation methodology is based on a **before-and-after comparison**, using historical data as a baseline.

The “before” scenario reflects the static pricing model based on fixed tables, while the “after” scenario simulates the application of price recommendations generated by the elasticity-based model.

Although the solution has not yet been fully deployed in production, the comparison is supported by:

- Historical sales and occupancy data;
- Counterfactual simulations using price variation scenarios;
- Expected behavioral changes inferred from elasticity coefficients.

2.3.2 Results and Impact Analysis:

Quantitative Impact

From the simulations and model outputs, the following impacts were identified:

- **Improved price alignment with demand:** Inelastic formats and periods show clear potential for price increases without significant loss of occupancy.
- **Reduction of inventory waste:** Elastic periods benefit from price reductions that increase sales volume and improve occupancy.
- **Revenue optimization potential:** Scenario analysis indicates that the model-suggested prices tend to be close to local revenue optima, outperforming uniform price increases or discounts.

While exact monetary gains depend on full operational deployment, the results indicate a **high potential for incremental revenue gains**, especially during peak-demand periods previously underpriced.

Qualitative Impact

Beyond financial outcomes, the solution delivers several qualitative benefits:

- **Improved decision-making:** Pricing decisions become data-driven rather than purely heuristic.
- **Operational agility:** Reduced dependence on manual price calculations and static tables.
- **Strategic pricing culture:** Pricing evolves from an operational task into a strategic management lever.
- **Transparency:** Clear rationale for price adjustments improves internal alignment between commercial and management teams.

2.3.3 Cost-Benefit Analysis:

The solution was developed using open-source technologies, minimizing direct financial costs. Estimated costs are primarily associated with:

- **Human resources:** data analysis, modeling, validation, and integration.

- **Integration effort:** connecting the model outputs to internal systems and dashboards.

No proprietary software licenses were required during development.

Return on Investment (ROI)

Given the infrastructure cost and the reuse of existing data, the solution presents a **favorable cost-benefit ratio**. Even modest improvements in price realization or occupancy rates would be sufficient to offset development costs.

For future production deployment, ROI can be formally calculated by comparing incremental revenue gains against integration and maintenance costs, with an expected short payback period due to the low marginal cost of running the models.

2.3.4 Critical Success Factors and Lessons Learned:

Critical Success Factors

- **Availability and quality of historical data:** Essential for robust modeling and reliable elasticity estimation.
- **Incremental experimentation:** Building models progressively helped isolate the impact of negotiation, clusters, and historical behavior.
- **Business alignment:** Continuous validation of assumptions with domain knowledge ensured practical relevance.
- **Model interpretability:** Using elasticity-based logic facilitated adoption by non-technical stakeholders.

Lessons Learned

- Pricing in the broadcasting market is heavily influenced by negotiation dynamics, not only by structural or seasonal factors.
- Predictive accuracy alone is insufficient; **diagnostic and prescriptive insights** are more valuable for pricing decisions.
- Occupancy optimization is substantially more complex than price estimation and requires additional data and modeling approaches.

- Early integration planning with backend systems is crucial to avoid manual data dependencies in production environments.

3 [Conclusion]

This project achieved its primary objective of designing and validating a data-driven dynamic pricing solution capable of supporting Rede Gazeta in the strategic optimization of advertising inventory and revenue.

Throughout the development, it became clear that traditional static pricing models are insufficient to capture the complexity of demand behavior in the broadcasting market. By combining historical sales data, capacity information, and econometric modeling, the project successfully transformed pricing from an operational rule-based process into a diagnostic and prescriptive decision-support mechanism.

The final elasticity-based model proved especially valuable, as it does not aim to predict a single “optimal” price, but rather to diagnose how demand reacts to price changes in different contexts. This approach enables the company to identify when prices can be safely increased to maximize margins, when reductions are necessary to stimulate volume, and when maintaining current prices is the most prudent strategy. As a result, pricing becomes a strategic lever aligned with both revenue maximization and inventory utilization.

From a business impact perspective, the solution provides Rede Gazeta with:

- Greater transparency over demand sensitivity across formats, periods, and durations;
- A structured framework to guide negotiations and pricing decisions;
- A scalable foundation for continuous improvement, supported by automated data pipelines and model retraining.

In addition, the project delivered a functional MVP that integrates model outputs into a frontend interface, demonstrating how analytical insights can be

operationalized and made accessible to commercial teams. This reinforces the practical applicability of the solution beyond academic experimentation.

Recommendations for Solution Evolution

To further enhance the value of the solution, the following next steps are recommended:

- **Integration with internal databases and live data sources**, replacing manual data inputs and enabling near real-time updates;
- **Daily-level data ingestion**, which would improve sensitivity to short-term demand fluctuations and increase model precision;
- **Expansion to other radio stations and regions**, leveraging the modular design of the pipeline;
- **Continuous monitoring and retraining**, ensuring that elasticity estimates remain aligned with evolving market behavior.

Knowledge Transfer and Maintenance

As part of the knowledge transfer plan, the project documentation details the data pipelines, modeling logic, assumptions, and limitations of the solution. This documentation enables Rede Gazeta's internal technical teams to maintain, adapt, and evolve the models independently. The use of widely adopted technologies such as Python and modular notebook-based pipelines further facilitates long-term sustainability and internal adoption.

In conclusion, the project not only met its technical and business objectives, but also established a robust analytical foundation for strategic pricing decisions, positioning Rede Gazeta to evolve toward a more data-driven and adaptive revenue management model.

References

VARIAN, Hal R. *Microeconomia: uma abordagem moderna*. 9. ed. Rio de Janeiro: Elsevier, 2015.

PHILLIPS, Robert. *Pricing and revenue optimization*. Stanford: Stanford University Press, 2005.

NAGLE, Thomas T.; HOGAN, John E.; ZALE, Joseph. *The strategy and tactics of pricing*. 6. ed. New York: Routledge, 2016.

MURPHY, Kevin P. *Machine learning: a probabilistic perspective*. Cambridge: MIT Press, 2012.

DAVENPORT, Thomas H.; HARRIS, Jeanne G. *Competing on analytics*. Boston: Harvard Business School Press, 2007.

PICARD, Robert G. *The economics and financing of media companies*. New York: Fordham University Press, 2011.

SHANKAR, Venkatesh; BOLTON, Ruth N. *An empirical analysis of determinants of retailer pricing strategy*. **Marketing Science**, v. 23, n. 1, p. 28–49, 2004.

PHILLIPS, Robert; LEE, Dennis. *Pricing models for media advertising inventory*. **Journal of Revenue and Pricing Management**, v. 17, n. 1, p. 23–39, 20

BREIMAN, Leo. *Random forests*. **Machine Learning**, v. 45, n. 1, p. 5–32, 2001.

CHEN, Tianqi; GUESTRIN, Carlos. *XGBoost: A scalable tree boosting system*. **Proceedings of the 22nd ACM SIGKDD**, p. 785–794, 2016.

BERTSIMAS, Dimitris; KALLUS, Nathan. *From predictive to prescriptive analytics*. **Management Science**, v. 66, n. 3, p. 1025–1044, 2020.

[LAST NAME, First Name. Article Title: Subtitle. **Journal Name**, Publication Location, Volume, Number, Pages, Publication Date. Available at: [electronic address]. Accessed on: [date of last access]]

Anexos